Theoretical Statistics

D.R. Cox and D.V. Hinkley

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PREFACE

This book is intended for statisticians wanting a fairly systematic development of the theory of statistics, placing main emphasis on general concepts rather than on mathematical rigour or on the detailed properties of particular techniques. The book is meant also to be suitable as a text-book for a second or third course in theoretical statistics. It is assumed that the reader is familiar with the main elementary statistical techniques and further has some appreciation of the way statistical methods are used in practice. Also knowledge of the elements of probability theory and of the standard special distributions is required, and we assume that the reader has studied separately the theory of the linear model. This is used repeatedly as an illustrative example, however.

We have reluctantly decided to exclude numerical examples and to attempt no detailed discussion of particular advanced techniques or of specific applications. To have covered these would have lengthened and changed the character of the book and to some extent would have broken the thread of the argument. However, in the training of students the working of set numerical exercises, the discussion of real applications and, if at all possible, involvement in current applied statistical work are of very great importance, so that this book is certainly not intended to be the basis of a self-contained course in statistics. To be quite specific, the more discursive parts of the book, for example on the usefulness and limitations of significance tests, will probably not be understandable without some experience of applications.

The mathematical level has been kept as elementary as possible, and many of the arguments are quite informal. The "theorem, proof" style of development has been avoided, and examples play a central role in the discussion. For instance, in the account of asymptotic theory in Chapter 9, we have tried to sketch the main results and to set out their usefulness and limitations. A careful account of the general theorems seems, however, more suited for a monograph than to a general book such as this.

Specialized topics such as time series analysis and multivariate

analysis are mentioned incidentally but are not covered systematically.

A major challenge in writing a book on theoretical statistics is that of keeping a strong link with applications. This is to some extent in conflict with the essential need for idealization and simplification in presentation and it is too much to hope that a satisfactory compromise has been reached in this book. There is some discussion of the connexion between theory and applications in Chapter 1.

The book deals primarily with the theory of statistical methods for the interpretation of scientific and technological data. There are applications, however, where statistical data are used for more or less mechanical decision making, for example in automatic control mechanisms, industrial acceptance sampling and communication systems. An introduction to statistical decision theory is therefore included as a final chapter.

References in the text have been restricted largely to those giving direct clarification or expansion of the discussion. At the end of each chapter a few general references are given. These are intended partly to indicate further reading, partly to give some brief historical background and partly to give some of the main sources of the material in the text. We felt that a very extensive bibliography would be out of place in a general book such as this.

At the end of every chapter except the first, there is an outline of some of the topics and results that it has not been feasible to include in the text. These serve also as exercises, although as such the level of difficulty is very variable and in some cases much detailed work and reference to original sources will be necessary.

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London, 1973

1 INTRODUCTION

1.1 Objectives of statistical analysis and theory

Statistical methods of analysis are intended to aid the interpretation of data that are subject to appreciable haphazard variability. The theory of statistics might then be expected to give a comprehensive basis for the analysis of such data, excluding only considerations specific to particular subject matter fields. In fact, however, the great majority of the theory, at any rate as presented in this book, is concerned with the following narrower matter.

There is chosen a family $\mathcal{F}$ of probability models, often completely specified except for a limited number of unknown parameters. From the data under analysis it is required to answer questions of one or both of the following types:

(a) Are the data consistent with the family $\mathcal{F}$?

(b) Assuming provisionally that the data are derived from one of the models in $\mathcal{F}$, what can be concluded about values of unknown parameters, or less commonly about the values of further observations drawn from the same model?

Problems (a) and (b) are related, but the distinction is a useful one. To a very large extent arithmetical rather than graphical methods of analysis are considered.

To illustrate the discussion consider the standard normal-theory model of simple linear regression. According to this, for data consisting of n pairs $(x_1, y_1), \ldots, (x_n, y_n)$, it is supposed that $y_1, \ldots, y_n$ correspond to random variables $Y_1, \ldots, Y_n$ independently normally distributed with constant variance σ^2 and with expected values

$$E(Y_j) = \gamma + \beta x_j \quad (j = 1, \ldots, n), \tag{1}$$

where γ and β are unknown parameters and $x_1, \ldots, x_n$ are regarded as known constants. This is a family $\mathcal{F}$ of models; a particular model is

obtained by specifying values for the parameters γ, β and σ^2. Often, but by no means always, primary interest would be in β.

The problem of type (a) would now be to examine the data for consistency with $\mathcal{F}$, some possibly important types of departure including non-linearity of the dependence of $E(Y_j)$ on x_j, non-constancy of $\text{var}(Y_j)$, lack of independence of the different Y_j's and non-normality of the distribution of the Y_j's. For problems of type (b) it would be assumed provisionally that the data are indeed derived from one of the models in $\mathcal{F}$ and questions such as the following would be considered:

Within what limits is it reasonable to suppose the parameter β to lie?

Are the data consistent with a particular theoretical value for, say, the parameter β?

Let $Y^\dagger$ be a further observation assumed to be obtained from the same model and parameter values as the original data, but with

$$E(Y^\dagger) = \gamma + \beta x^\dagger.$$

Within what limits is $Y^\dagger$ expected to lie?

In the theoretical discussion, it will be usual to take a family of models as given. The task is to formulate meaningful and precise questions about such models and then to develop methods of analysis that will use the data to answer these questions in a sensitive way.

In more detail, in order to deal with the first of these questions just mentioned, the first step is to formulate a precise meaning for such limits for β; this is done in the concept of confidence limits (Section 7.2). Then, there usually being many ways of computing confidence limits, at least approximately, the next step is to define criteria giving the best such limits, or failing that, at least reasonably good confidence limits. Finally, general constructive procedures are required for computing good confidence limits for specific problems. This will lead to sensitive procedures for the analysis of data, assuming that the family $\mathcal{F}$ is well chosen. For example, in the special linear regression model (1), we would be led to a procedure for computing from data limits for, say, the parameter β which will in a certain sense give the most precise analysis of the data given the family $\mathcal{F}$.

In this, and corresponding more general theoretical discussions, much emphasis is placed on finding optimal or near optimal procedures within a family of models.

How does all this correspond with the problems of applied statistics? The following points are relevant:

(i) The choice of the family $\mathcal{F}$ of models is central. It serves first to define the primary quantities of interest, for example, possibly the parameter β in (1), and also any secondary aspects of the system necessary to complete the description. Some of the general considerations involved in the choice of $\mathcal{F}$ are considered in Section 1.2.

(ii) Except occasionally in very simple problems the initial choice of $\mathcal{F}$ will be made after preliminary and graphical analysis of the data. Furthermore, it will often be necessary to proceed iteratively. The results of analysis in terms of a model $\mathcal{F}$ may indicate a different family which may either be more realistic or may enable the conclusions to be expressed more incisively.

(iii) When a plausible and apparently well-fitting family is available it is attractive and sometimes important to use techniques of analysis that are optimal or nearly so, particularly when data are rather limited. However, criteria of optimality must always be viewed critically. Some are very convincing, others much less so; see, for example, Sections 5.2(iii) and 8.2. More importantly, it would be poor to use an analysis optimal for a family $\mathcal{F}$ if under an undetectably different family $\mathcal{F}'$ the same analysis is very inefficient. A procedure that is reasonably good for a broad range of assumptions is usually preferable to one that is optimal under very restrictive assumptions and poor otherwise. Thus it is essential to have techniques not only for obtaining optimal methods but also for assessing the performance of these and other methods of analysis under non-standard conditions.

(iv) At the end of an analysis, it is always wise to consider, even if only briefly and qualitatively, how the general conclusions would be affected by departures from the family $\mathcal{F}$. Often this is conveniently done by asking questions such as: how great would a particular type of departure from $\mathcal{F}$ have to be for the major conclusions of the analysis to be altered?

(v) It is important to adopt formulations such that the statistical analysis bears as directly as possible on the scientific or technological meaning of the data, the relation with previous similar work, the connexion with any theories that may be available and with any practical application that may be contemplated. Nevertheless, often the statistical analysis is just a preliminary to the discussion of the underlying meaning of the data.

Some of these points can be illustrated briefly by the regression

problem outlined above.

The minimal preliminary analysis is a plot of the points (x_j, y_j) on a scatter diagram. This would indicate, for example, whether a transformation of the variables would be wise before analysing in terms of the model (1) and whether there are isolated very discrepant observations whose inclusion or exclusion needs special consideration. After an analysis in terms of (1), residuals, i.e. differences between observed values and estimated values using the model (1), would be calculated, and analysis of these, either graphically or arithmetically, might then suggest a different family of models. With a more complex situation, an initial family of models may be complicated and one might hope to be able to pass to a simpler family, for example one in which there were appreciably fewer unknown parameters. In the present case, however, any change is likely to be in the direction of a more complex model; for instance it may be appropriate to allow $\mathrm{var}(Y_j)$ to be a function of x_j.

It might be argued that by starting with a very complex model, some of the difficulties of (ii)–(iv) could be avoided. Thus one might set up a very general family of models involving additional unknown parameters describing, for example, the transformations of the x- and y-scales that are desirable, the change with x of $\mathrm{var}(Y)$, the lack of independence of different observations and non-normality of the distributions. The best fitting model within this very general family could then be estimated.

While this type of approach is sometimes useful, there are two serious objections to it. First, it would make very heavy going of the analysis even of sets of data of simple structure. A more fundamental reason that a single formulation of very complex models is not in general feasible is the following.

In designing an experiment or scheme of observations, it is important to review beforehand the possible types of observations that can occur, so that difficulties of analysis and ambiguities of interpretation can be avoided by appropriate design. Yet experience suggests that with extensive data there will always be important unanticipated features. An approach to the analysis of data that confines us to questions and a model laid down in advance would be seriously inhibiting. Any family of models is to be regarded as an essentially tentative basis for the analysis of the data.

The whole question of the formulation of families of models in the light of the data is difficult and we shall return to it from time

to time.

To sum up, the problems we shall discuss are closely related to those of applied statistics but it is very important to ensure that the idealization which is inevitable in formulating theory is not allowed to mislead us in practical work.

1.2 Criteria for the choice of families of models

In the previous section the importance has been explained of choosing a family $\mathcal{F}$ of probabilistic models in terms of which questions are to be formulated and methods of statistical analysis derived. For the more elaborate parts of statistical theory we start from the representation of observations in terms of random variables and the idea that normally the parameters of the underlying probability distributions are the quantities of real interest. Yet this view needs to be treated with proper caution and should not be taken for granted. Where the data are obtained by the random sampling of a physically existing population, the parameters have a reasonably clear meaning as properties of the population. In other cases the probability distributions refer to what would happen if the experiment were repeated a large number of times under identical conditions; this is always to some extent hypothetical and with "unique" data, such as economic time series, repetition under identical conditions is entirely hypothetical. Nevertheless the introduction of probability distributions does seem a fruitful way of trying to separate the meaningful from the accidental features of data. Parameters are to be regarded sometimes as representing underlying properties of a random system and sometimes as giving concise descriptions of observations that would be obtained by repetition under the same conditions.

The methods of most practical value are those that combine simple description of the data with efficient assessment of the information available about unknown parameters.

It is hard to lay down precise rules for the choice of the family of models, but we now list briefly some of the considerations involved. These include:

(a) The family should if possible usually establish a link with any theoretical knowledge about the system and with previous experimental work.

(b) There should be consistency with known limiting behaviour. For example, it may be known or strongly suspected that a curve

approaches an asymptote or passes through some fixed point such as the origin. Then, even though this limiting behaviour may be far from the region directly covered by the data, it will often be wise to use a family of models consistent with the limiting behaviour or, at least, to use a family reducing to the limiting behaviour for special parameter values.

(c) So far as possible the models should be put in a form such that the different parameters in the model have individually clear-cut general interpretations.

(d) Further, a description of the data containing few parameters is preferable. This may involve looking at a number of different families to find the most parsimonious representation. There is some chance of conflict with requirements (a) and (b). Indeed in some cases two different analyses may be desirable, one bringing out the link with some relevant theory, the other expressing things in their most economical form.

(e) It is, of course, desirable that the statistical theory associated with the family of models should be as simple as possible.

A fairly recent trend in statistical work places some emphasis on the construction of special models, often by constructing stochastic processes representing the system under investigation. To be at all realistic these often have to be extremely complex. Indeed they may contain so many adjustable parameters that little can be learnt by fitting them to any but extremely extensive data. It is therefore worth stressing that very simple theoretical analyses can be valuable as a basis for statistical analysis. We then choose a family reducing to the theoretical model for special values of certain unknown parameters. In that way we can find how much of the variation present can be accounted for by the simple theory and also characterize the departures from the theory that may be present. It is not at all necessary for a theory to be a perfect fit for it to be a useful tool in analysis.

It will be seen later that the statistical theory of a family is simplified appreciably whenever the family is of the exponential type; see Section 2.2(vi). This provides a fairly flexible set of models and it will often be best to start with a model of this type; for example, nearly all the procedures of multivariate analysis are closely associated with the family of multivariate normal models. The effect of departures can then be considered. It is, however, too much to hope that a reasonably adequate model of exponential type can always be found without violating some other more important requirement.

The requirement of simplicity of statistical analysis may well be in conflict with one or more of the other requirements, for example (a) and (b). Thus the widely used procedure of fitting polynomial curves and surfaces leads to representations whose limiting behaviour is usually unreasonable. This may not matter where the polynomials are used in a limited way for interpolation but seems often likely to be a severe restriction on their usefulness in the interpretation of data; exceptions are when a small quadratic term is used to assess the direction and magnitude of the departure from an expected linear relationship, and when local behaviour in the neighbourhood of a stationary value is under investigation.

1.3 The analysis of complex responses

A widely occurring problem in applied statistics, particularly where automatic recording equipment is used, is the analysis of data in which the response of each individual (subject, animal, batch of material, etc.) is complex and, for example, may consist of a long sequence of observations possibly of several types. Thus in some types of psychological experiment, there will be for each subject a response consisting of a long sequence of successes and failures at some task. In all there may be several such trials for each of many subjects, the whole covering a number of experimental treatments.

It may sometimes be feasible to formulate a single family of models that will embrace all aspects of the data, but more commonly it will be wise to proceed in several steps. First we try to summarize each complex response in a small number of quantities. These may be derived from some formal theoretical analysis, for instance from a simplified mathematical theory of the phenomenon. Or again the formal techniques of time series analysis and multivariate analysis are often guides to the choice of summarizing values. Finally such quantities may be derived from a qualitative inspection and graphical analysis of a sample of individual responses. That is, we construct a few quantities thought to summarize interesting features of the response, such as, for example, average level, trend, amount and nature of local variability. With very complex data it may take much investigation to find the best quantities for further analysis.

Then at the second stage of the analysis the output of the first stage is used as the input data to estimate treatment effects, etc., often by a linear model. There is the following important implication

for the discussion in the rest of this book: the observations for analysis and for which we set up a probability model may be the original data or they may be quantities derived from the preliminary analysis of more complex data. This fact gives the relatively simple models that we shall discuss much wider usefulness than would otherwise be the case.

1.4 Plan of the book

Chapter 2 introduces some fundamental ideas about likelihood and sufficiency which are central to much of the subsequent work. The second part of the chapter, which can be omitted on a first reading, compares some of the broad general approaches to statistical inference.

Chapters 3–7, which in some ways are the core of the book, deal with significance testing and with interval estimation. Chapter 8 is concerned with point estimation. Chapter 9 is on asymptotic theory. For large samples this gives approximate solutions to problems for which no simple "exact" solutions are possible by the techniques of Chapters 3–7.

Finally the last two chapters deal with procedures based on the availability of a prior probability distribution for the unknown parameters of the family $\mathcal{F}$, and with decision theory.

To some extent, the chapters can be taken in a different order. For example, some readers may want to take the rather introductory Chapters 10 and 11 relatively early.

It would be very restricting to use throughout a single completely uniform notation and it is virtually impossible to avoid use of the same letter for different things in different contexts. However, as far as is practicable, the following conventions have been followed.

Random variables representing observations or functions calculated from observations are denoted by capital letters such as X, Y and Z. For example, Y denotes the set of random variables $Y_1, \ldots, Y_n$ considered as a column vector. On the whole Y is reserved for the observation under analysis. To distinguish between random variables and the particular realizations of them forming the data under analysis, the observed values will be denoted by corresponding lower case letters, for example y. It is, however, occasionally convenient to be inconsistent and to retain the capital letters for the observations, where there is no danger of confusion.

Random variables that are not observable, for example "errors", are sometimes denoted by the Greek letters ϵ and η. The Greek letter α is almost exclusively reserved for certain probabilities arising in the study of tests and interval estimation.

Fixed constants are denoted by lower case letters $a, b, c, \ldots$

Unknown parameters are usually denoted by the Greek letters θ, $\phi, \ldots$ They may be scalars or vectors depending on the context; when vectors they are to be considered as column vectors. In dealing with particular situations standard symbols are used; for example, the mean and variance of a normal distribution are denoted by μ and σ^2.

For a random variable, say U, the cumulative distribution function is denoted by $F_U(x)$, or by $F_U(x\,;\theta)$ if it is required to stress dependence on a parameter θ. That is, $F_U(x) = \mathrm{pr}(U \leqslant x)$, where $\mathrm{pr}(A)$ denotes the probability of the event A. The corresponding probability density function, a term used for both continuous and discrete random variables, is denoted by $f_U(x)$ or $f_U(x\,;\theta)$. Thus if U has a Poisson distribution of mean μ, then

$$f_U(x\,;\mu) = \frac{e^{-\mu}\mu^x}{x!} \quad (x = 0, 1, 2, \ldots),$$

whereas if V has a normal distribution of mean μ and variance σ^2, written $N(\mu, \sigma^2)$, then

$$f_V(x\,;\mu, \sigma) = \frac{1}{\sqrt{(2\pi)}\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}.$$

The p dimensional multivariate normal distribution of mean the column vector μ and covariance matrix $\boldsymbol{\Sigma}$ is denoted by $MN_p(\mu, \boldsymbol{\Sigma})$. A direct extension of the notation for densities is used for joint and conditional densities.

Standard notation is used for expectations, variances and covariances. Where it is required to show the parameter value under which, say, an expectation is calculated, we write, for example, $E(Y\,;\theta)$.

Bold fount is restricted to matrices; the transpose of a matrix or vector is denoted by a superscript T.

An estimate of a parameter θ will sometimes be denoted $\tilde{\theta}$; the notation $\hat{\theta}$ will be restricted to maximum likelihood and least squares estimates. All such estimates $\tilde{\theta}$ are functions of the data, i.e. correspond to random variables. Therefore if the general conventions were to be followed a capital letter would be used for the random variable, but this is often inconvenient. Sums of squares and mean squares

arising in linear model analyses have been denoted SS and MS, sometimes with an appropriate suffix, it being clear from the context whether random variables or observed values are involved.

Asterisks are very largely reserved for tabulated constants arising in significance tests, etc. By convention we use values corresponding to an upper tail area. For example, k_α^* denotes the upper α point of the standard normal distribution, i.e. $\Phi(k_\alpha^*) = 1 - \alpha$, where $\Phi(.)$ is the standard normal integral.

Abbreviations have been kept to a minimum. The only ones widely used are i.i.d. (independent and identically distributed), p.d.f. (probability density function), referring to both continuous and discrete random variables, and m.l.e. (maximum likelihood estimate).

Bibliographic notes

Discussion of the relation between theory and application is most often found in reports of general lectures. Presidential addresses to the Royal Statistical Society by Fisher (1953), Pearson (1956), Kendall (1961), Bartlett (1967), Yates (1968) and Barnard (1972) all in part bear on this issue; see also Kempthorne (1966). Neyman (1960) emphasizes the role of special stochastic models. Tukey and Wilk (1966) have argued in favour of an increased emphasis on graphical methods and on descriptive statistics generally, i.e. on methods which are not based on explicit probabilistic arguments.

Of the many books on statistical methods, that of Daniel and Wood (1971) particularly well illustrates the interplay between the analysis of data and the choice of model. Of those on theory, the later chapters of Kempthorne and Folks (1971) give an introduction to many of the topics of the present book from a viewpoint similar to that taken here. Silvey (1970) gives a concise mathematical introduction. The comprehensive book of Rao (1973) emphasizes distribution theory and the linear model. The three volumes of Kendall and Stuart (1967–69) are particularly valuable in providing an introduction to a wide range of special topics.

A connected account of the history of statistical inference is not available at the time of writing. A collection of papers on various historical topics is edited by Pearson and Kendall (1970); there are a number of biographical articles in *International Encyclopedia of Social Sciences* (Sills, 1968).